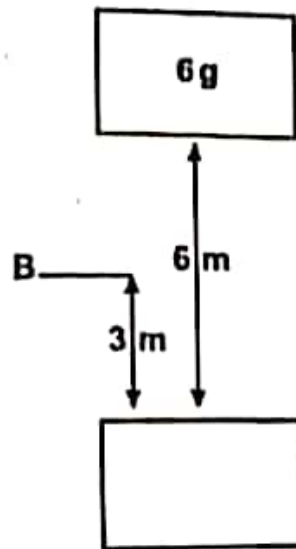


# Physics

## Physical Quantities and Units

- Which one is the highest power multiple?  
 A) Giga  
 C) Tera✓  
 B) Mega  
 D) Deca
- SI unit of charge is \_\_\_\_\_.  
 A) Ampere✓  
 C) Volt  
 B) Coulomb  
 D) Calorie
- The electrical analog of mass is electricity is \_\_\_\_\_.  
 A) Capacitance  
 C) Inductance  
 B) Charge✓  
 D) Resistance
- Which one of the following relations is correct?  
 A)  $1 \text{ wb m}^{-2} = \text{N m}^{-1} \text{ A}^{-1}$   
 C)  $1 \text{ Tesla} = 10^4 \text{ Gauss}$   
 B)  $1 \text{ wb m}^{-2} = 1 \text{ Tesla}$   
 D) All of these✓
- Which one of the following relations is correct?  
 A)  $1 \text{ wb m}^{-2} = \text{N m}^{-1} \text{ A}^{-1}$   
 B)  $1 \text{ wb m}^{-2} = 1 \text{ Tesla}$   
 C)  $1 \text{ Tesla} = 10^4 \text{ Gauss}$   
 D) All of these✓
- The principle of homogeneity of dimensions determines  
 A) Only variable in the equation  
 B) Correctness of an equation✓  
 C) Only constant in the equation  
 D) Constant and variable in the equation
- The dimensions of gravitational constant "G" are:  
 A)  $[ML^{-2}T^{-1}]$   
 C)  $[M^2L^{-2}T^{-1}]$   
 B)  $[ML^{-2}T^{-2}]$   
 D)  $[M^{-1}L^3T^{-1}]$ ✓
- Ultraviolet radiations cause:  
 A) Severe Crop Damage  
 C) Sunburn, blindness, skin cancer  
 B) Decay of Microorganisms  
 D) All of the above✓
- Unit vector in the direction of vector  $2\hat{i} - 4\hat{j}$  will be:  
 A)  $\frac{2\hat{i} - 4\hat{j}}{\sqrt{6}}$   
 C)  $\frac{4\hat{i} - 2\hat{j}}{\sqrt{10}}$   
 B)  $\frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$ ✓  
 D)  $\frac{\hat{i} - 2\hat{j}}{\sqrt{7}}$
- If the force of magnitude 8 N acts on a body in direction making an angle 30, its X and Y components will be:  
 A)  $F_x = 3\sqrt{3}$   $F_y = 4$   
 C)  $F_x = 4\sqrt{3}$   $F_y = 4$ ✓  
 B)  $F_x = 4$   $F_y = 8$   
 D)  $F_x = 8$   $F_y = 4\sqrt{3}$
- The scalar product of i and k is:  
 A) Zero✓  
 C) 90°  
 B) 1  
 D) -1

Q.12 A body of mass 6 g falls under action of gravity. At initial position 'A' its P.E. is 480 J and K.E. is 0 J. During its downward journey at point 'B' its energies will be ( $g = 10 \text{ ms}^{-2}$ ):



- A) P.E. = 300 J and K.E. = 180 J ✓  
 B) P.E. = 230 J and K.E. = 250 J

- C) P.E. = 180 J and K.E. = 300 J  
 D) P.E. = 250 J and K.E. = 230 J

Q.13 If the mass of the bob of a pendulum is doubled its time period is:

- A) Halved.  
 C) Doubled.

- B) Unchanged. ✓  
 D) Increases four times.

Q.14 Force in terms of base units is expressed as

- A)  $\text{kg ms}^{-2}$ . ✓  
 C)  $\text{kg m}^2 \text{s}^{-2}$ .

- B)  $\text{kg m}^2 \text{s}^{-3}$ .  
 D) None of these.

Q.15 The units of E in  $E=mc^2$  are

- A)  $\text{kg m s}^{-2}$ .  
 C)  $\text{N m s}^{-2}$ .

- B)  $\text{kg m}^2 \text{s}^{-2}$ . ✓  
 D) Both B and C.

Q.16 Light year is a measure of

- A) Distance. ✓  
 C) Time.

- B) Intensity of light.  
 D) Velocity.

Q.17 When the dimensions of both sides of an equation are equal, then the equation is said to be

- A) Simultaneous  
 C) Homologous ✓

- B) Instantaneous  
 D) Quadratic

Q.18 Radian is a unit of angular displacement which can also be measured in degrees. How many radians are equal to one degree?

- A)  $\frac{180}{\pi}$

- B)  $\frac{2\pi}{180}$

- C)  $\frac{\pi}{180}$  ✓

- D)  $\frac{\pi}{57.3}$

Q.19 An electric charge in uniform motion produces:

- A) An electric field.  
 C) A magnetic field.

- B) Both magnetic and electric fields. ✓  
 D) Neither magnetic nor electric fields.

If 'm' is the mass, 'c' is the velocity of light and  $x = mc^2$ , then dimensions of 'x' will be:

- A)  $[LT^{-1}]$   
 B)  $[MLT^{-1}]$   
 C)  $[ML^2T^{-2}]$  ✓  
 D)  $[MLT^{-2}]$

The wavelength ' $\lambda$ ' of a wave depends on the speed ' $v$ ' of the wave and its frequency ' $f$ '. Decide which of the following is correct?

- A)  $f = v\lambda$   
 B)  $f = \frac{v}{\lambda}$  ✓  
 C)  $f = \frac{\lambda}{v}$   
 D)  $f = v\lambda^{-2}$

Name the quantity which can be measured by using base unit ' $kgm^2s^{-3}$ '

- A) Weight  
 B) Power  
 C) Pressure  
 D) Work ✓

The formula for electric field strength is ' $E = F/Q$ ', where E is electric field strength and F is force and Q is charge. Which one of the following options gives the correct base units for electric field strength?

- A)  $kgms^{-3}A^{-1}$  ✓  
 B)  $kg^2m^{-2}s^{-3}A$   
 C)  $kg s^{-2}A^{-3}$   
 D)  $ms^{-1}A^{-3}$

Which set of the prefixes gives values in increasing order?

- A) Pico, Mega, Kilo, Tera  
 B) Tera, Pico, Micro, Kilo  
 C) Pico, Micro, Mega, Giga ✓  
 D) Giga, Kilo, Milli, Nano

The unit of temperature in base unit is:

- A) Celsius  
 B) Kelvin ✓  
 C) Degree  
 D) Fahrenheit

## Forces

The horizontal range of a projectile, at a certain place, is completely determined by

- A) The angle of projection ✓  
 B) The mass of the projectile  
 C) The initial velocity of projection  
 D) Speed and mass of the projectile

If velocity is double, then.

- A) Momentum increase 4 times and K.E increases 2 times  
 C) Momentum and K.E remain same  
 B) Momentum increases 2 times and K.E increase constant  
 D) Momentum increases 2 times and K.E increases 4 times ✓

For a body to be in complete equilibrium

- A) Linear acceleration is zero  
 C) Angular acceleration is zero  
 B) Linear acceleration is zero but angular acceleration is not zero  
 D) Linear acceleration and angular acceleration both should be zero ✓

If a force of magnitude 8 N acts on a body in direction making an angle 30, its x and y components are

- A)  $F_x = 4\sqrt{3}$  and  $F_y = 8$   
 B)  $F_x = 4\sqrt{3}$  and  $F_y = 4$  ✓



- Q.5 The difference of a vector  $\vec{B}$  and its negative vector  $-\vec{B}$  is
- A) A null vector  
 C) Equal to magnitude of vector  $\vec{B}$   
 B) Twice the magnitude of vector  $\vec{B}$  ✓  
 D) Smaller than magnitude of vector  $\vec{B}$
- Q.6 Time of projectile's flight is
- A)  $\frac{v_i^2 \sin^2 \theta}{g}$   
 B)  $\frac{v_i^2 \sin \theta}{g}$   
 C)  $\frac{2v_i \sin \theta}{g}$  ✓  
 D)  $\frac{v_i^2}{g} \sin \theta$
- Q.7 If the velocity of the body changes by equal amount in equal intervals of time, the body is said to have:
- A) variable acceleration  
 B) uniform velocity  
 C) uniform acceleration ✓  
 D) negative acceleration
- Q.8 In order to determine the maximum height of the projectile, the equation of motion used is
- A)  $aS = v_f^2 - v_i^2$   
 B)  $2S = a(v_f^2 - v_i^2)$   
 C)  $2aS = v_f^2 - v_i^2$  ✓  
 D)  $aS = (v_f^2 - v_i^2)$
- Q.9 If a force of 12 N acts on a car and changes its momentum from 36 kgm/sec to 60 kgm/sec, the time during which this change occurs will be
- A) 24 sec  
 B) 12 sec  
 C) 2 sec ✓  
 D) 8 sec
- Q.10 Which one of the following is a non-conservative force?
- A) Electric force  
 B) Gravitational force  
 C) Elastic spring force  
 D) Frictional force ✓
- Q.11 Value of escape velocity for the surface of the earth is 11 km/sec. Its value for surface of the moon is
- A) 11 km/sec  
 B) 2.4 km/sec ✓  
 C) 10.4 km/sec  
 D) 4.3 km/sec
- Q.12 On a clear day at noon, the intensity of solar energy reaching the earth's surface is about
- A)  $1.0 \text{ kWm}^{-2}$  ✓  
 B)  $1.0 \text{ Wm}^{-2}$
- Q.13 When a lift is accelerated upward, the apparent weight of an object in it will be
- A) Equal to its real weight  
 B) Zero  
 C) Less than its real weight  
 D) Greater than its real weight ✓
- Q.14 The moment of inertial of a thin rod is
- A)  $\frac{1}{2} mL^2$   
 B)  $\frac{1}{12} mL^2$

C)  $\frac{1}{4} \text{ m}^3 \text{ L}$

D)  $\frac{1}{12} \text{ mL}^2 \checkmark$

- Q.15** In elastic collision, when a massive body collides with light body at conditions  $m_1 \gg m_2$  and  $v_2 = 0 \text{ ms}^{-1}$ , then the change in velocity will be written as:
- A)  $v_1' \approx -v_1$ ;  $v_2' \approx v_1$       B)  $v_1' \approx v_1$ ;  $v_2' \approx 2v_1 \checkmark$   
 C)  $v_1' \approx v_1$ ;  $v_2' \approx 0$       D)  $v_1' \approx -v_1$ ;  $v_2' \approx 0$

## Fluid Dynamics

- Q.1** Conservation of mass of fluid flow leads to  
 A) Bernoulli's equation      B) Equation of motion  
 C) Venturi meter      D) Equation of continuity  $\checkmark$
- Q.2** The blood vessels collapse when  
 A) External pressure applied becomes greater than the systolic pressure  $\checkmark$   
 C) External pressure applied is equal to systolic pressure  
 B) External pressure applied is less than the systolic pressure  
 D) External pressure applied is zero
- Q.3** In Millikan's Method, the radius of droplet can be calculated by:  
 A)  $r = \sqrt{\frac{qv_t}{2\rho g}}$       B)  $r^2 = \frac{9\eta v_t}{2\rho g} \checkmark$   
 C)  $r^2 = \frac{q\eta v_t}{\rho g}$       D)  $r = \frac{9\eta v_t}{2\rho g}$
- Q.4** A tiny droplet of oil of density ' $\rho$ ' and radius ' $r$ ' falls through air under force of gravity. If viscosity of air is ' $\eta$ ', the terminal velocity acquired by the oil drop is given by:  
 A)  $v_t = \frac{4gr^2\rho}{9\eta}$       B)  $v_t = \frac{2gr^2\rho}{9\eta} \checkmark$   
 C)  $v_t = \frac{9\eta r^2\rho}{4g}$       D)  $v_t = \frac{9\eta r^2\rho}{2g}$
- Q.5** Torricelli's theorem be written as:  
 A)  $v_2 = \sqrt{2g(h_1 - h_2)} \checkmark$       B)  $v_2 = \sqrt{2g(h_2 - h_1)}$   
 C)  $v_2 = \sqrt{g(h_2 - h_1)}$       D)  $v_2 = \sqrt{g(h_1 - h_2)}$
- Q.6** A pipe varies uniformly in diameter from 2 m to 4 m. An incompressible fluid enters the pipe with velocity 16m/sec. What is velocity of fluid when it leaves the pipe?  
 A) 64 m/sec.      B) 8 m/sec.  
 C) 32 m/sec.      D) 4 m/sec.  $\checkmark$
- Q.7** The drag force decreases as the speed of an object moving through fluid \_\_\_\_\_  
 A) Increases.      B) Remains constant.  
 C) Decreases.  $\checkmark$       D) Both B and C.
- Q.8** An object having spherical shape of radius ' $r$ ' experiences a retarding force  $F$  from a

fluid of co-efficient of viscosity ' $\eta$ ' when moving through the fluid with speed ' $v$ '.  
What is the ratio of retarding force to speed?

A)  $6\pi\eta r^2$

C)  $6\pi\eta/r^2$ ✓

B)  $6\pi\eta r$

D)  $6\pi\eta/r$

- Q.9 When the drag force is equal to the weight of the droplet, the droplet will fall with:  
A) High Speed  
B) Certain acceleration  
C) Low Speed  
D) Constant Speed✓

- Q.10 The density of blood is:  
A) Less than water  
B) Greater than water  
C) Nearly equal to water✓  
D) Three times that of water

- Q.11 Stokes' Law for steady motion in a fluid of infinite extent is given by  
A)  $F = 6\pi\eta rv$ ✓  
B)  $F = 6\pi\eta r^2\rho$   
C)  $F = (4/3)\pi r^3\rho g$   
D)  $F = 2gr^2\rho/9\eta$

- Q.12 If speed of efflux through a small hole in a large tank is 9.8 m/s. Find the height at the fluid above the hole

A) 1 m

C) 9.8 m

B) 4.9 m✓

D) 19.6 m

- Q.13 Flow speed of the fluid through a non-uniform pipe increases from 1 m/sec to 3 m/sec. If change in P.E. is zero, then pressure difference between two points will be: (density of the fluid =  $1000 \text{ kg/m}^3$ )

A)  $1000 \text{ N/m}^2$

C)  $9000 \text{ N/m}^2$

B)  $8000 \text{ N/m}^2$

D)  $4000 \text{ N/m}^2$ ✓

- Q.14 Stokes' Law is given as:

A)  $F = 6\pi\eta r^2 v$

C)  $F = 6\pi\eta rv$ ✓

B)  $F = 6\pi\eta rv^{-1}$

D)  $F = 6\pi^2\eta r^3 v$

- Q.15 The product of cross-sectional area of the pipe and the fluid speed at any point along the pipe:

A) Remains constant✓

C) Is zero

B) Exponentially increases

D) Exponentially decreases

- Q.16 A small leak is developed in a large water storage tank. If the height of water above leakage is 10 m, then find the speed of efflux through the leak:

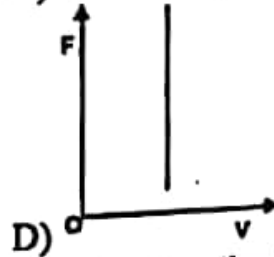
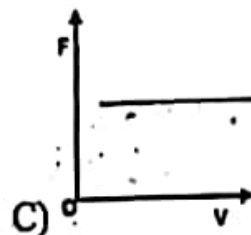
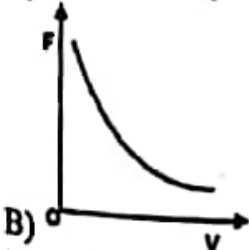
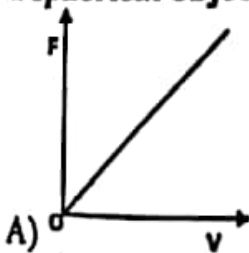
A) 14 m/sec✓

C) 10 m/sec

B) 9.8 m/sec

D) 20 m/sec

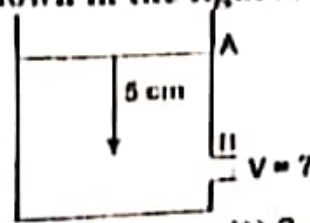
- Q.17 Which of the following is the best graphical representation between drag force ' $F$ ' on a spherical object of radius ' $r$ ' and its speed ' $v$ ' through a fluid of viscosity ' $\eta$ '?



- Q.18 What is the speed of an incompressible non-viscous liquid flowing out from 'B'?



contained in a container as shown in the figure? Where  $AB = 5 \text{ m}$  and  $g = 10 \text{ m/s}^2$ .



- A) 5 m/s  
C) 10 m/s✓

- B) 2 m/s  
D) 50 m/s

Q.19 For the horizontal pipe, the fluid inside it is flowing horizontally then Bernoulli's equation can be written as

- A)  $P + \rho v^2 = \text{constant}$   
C)  $2P + \rho v^2 = \text{constant}$ ✓

- B)  $P + 2\rho v^2 = \text{constant}$   
D)  $2P + 2\rho v^2 = \text{constant}$

Q.20 In fluid flow, for the equation of continuity  $A_1 v_1 = A_2 v_2$ . If velocity of the fluid at one end is doubled, then what will be the cross-sectional area at this end?

- A) Double  
C) Half✓

- B)  $(\text{Half})^2$   
D)  $(\text{Double})^2$

Q.21 Mass flow per second of the fluid is given by:

## Light

Q.1 In Newton ring apparatus, at the point of contact of the lens and glass plate, the additional path difference introduced is

- A)  $\lambda/4$   
C)  $\lambda/2$ ✓

- B)  $\lambda$   
D)  $\lambda/3$

Q.2 In the case of a grating spectrometer, the resolving power 'R' of the grating is defined as

- A)  $\lambda / \Delta\lambda$ ✓  
C)  $\lambda / D$

- B)  $\lambda / \lambda_1$   
D)  $N \times m$

Q.3 Which one of the following lights travels fastest in optical fibers?

- A) Visible light  
C) Ultraviolet light

- B) Ordinary light  
D) Invisible infrared light✓

Q.4 In Young's Double Slit Experiment, if the distance between slits and screen is doubled, then fringe spacing becomes:

- A) Zero  
C) One

- B) Doubles of the original value✓  
D) Half of the original value

Q.5 In Michelson's interferometer 792 bright fringes pass across the field of view when its movable mirror is displaced through 0.233 mm using the equation  $I = m \frac{\lambda}{2}$  the

wavelength of light used is:

- A) 588 nm✓  
C) 620 nm

- B) 348 nm  
D) 400 nm

Q.6 In Michelson's Experiment, the formula to calculate the speed of light is:

- A)  $c = 2fd$

- B)  $c = \frac{16f}{d}$

$$C) c = \frac{2\lambda}{d}$$

$$D) c = 16 \text{ fd} \checkmark$$

Q.7 The information received at the other end of a fibre can be inaccurate due to \_\_\_\_\_ of the light signal.

- A) Longer wavelengths  
C) Frequency

- B) Intensity  
D) Dispersion or Spreading  $\checkmark$

Q.8 The centre of Newton rings is dark due to:

- A) Polarization.  
C) Destructive interference.  $\checkmark$

- B) Constructive interference.  
D) Reflection.

Q.9 Which of the following lights travels the fastest in optical fibres?

- A) Visible light.  
C) Invisible infra-red.  $\checkmark$

- B) Ultra-violet.  
D) Ordinary light.

Q.10 A single mode step index fibre has core of about \_\_\_\_\_  $\mu\text{m}$  diameter

- A) 50 to 1000.  
C) 50. D) 5.  $\checkmark$

- B) 30.

Q.11 A watch maker uses \_\_\_\_\_ to repair the watches.

- A) Telescope.  
C) Convex mirror.

- B) Convex lens.  $\checkmark$   
D) Concave lens.

Q.12 The ratio of the \_\_\_\_\_ is called magnification.

- A) Image size to object size.  $\checkmark$   
C) Object size to image size.

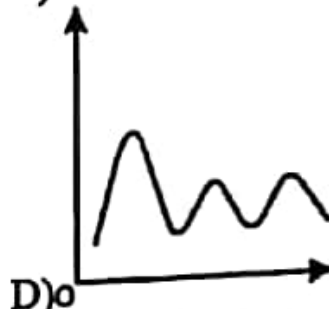
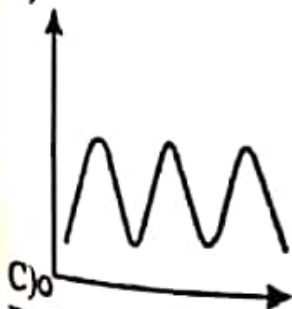
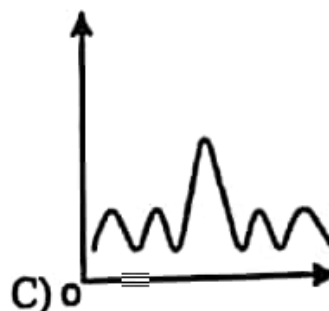
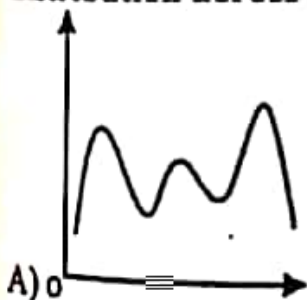
- B) Eyepiece size to object size.  
D) None of these

Q.13 A yellow light of wavelength 500 nm emitted by a single source passes through two narrow slits 1 mm apart. How far apart are two adjacent bright fringes when interference is observed on a screen 10 m away?

- A) 5 mm.  $\checkmark$   
C) 1.33 mm.

- B) 0.5 mm.  
D) 50 mm.

Q.14 A monochromatic light of wavelength ' $\lambda$ ' is used to produce the diffraction pattern through a single slit of width X. Which one of the following represents the intensity distribution across the screen?



Q.15 For interference of light waves to take place, the required condition is

- A) The path difference of the light waves from the two sources must be large  
C) The interfering waves must be non-coherent



- B) The light waves may come from different sources  
D) The light waves must come from two coherent sources✓
- Q.16 The property of bending of light around an obstacle and spreading of light wave into geometric shadow of an obstacle is called:  
A) Diffraction of Light✓  
B) Quantization of Light  
C) Polarization of Light  
D) Interference of Light
- Q.17 The normal human eye can focus a sharp image of an object on the eye if the object is located at certain distance called  
A) Least Point  
B) Far Point  
C) Near Point✓  
D) Distinct Point
- Q.18 Polarization of light exhibited the nature of light as  
A) Longitudinal wave  
B) Transverse wave✓  
C) Compressional wave  
D) Electromagnetic wave
- Q.19 The concentration of a sugar solution can be determined by  
A) Un-polarized light  
B) Interference of light  
C) Plane polarized light✓  
D) Diffraction of light
- Q.20 The information from one place to another can be transmitted very safely and easily by:  
A) Copper wire  
B) Photodiode  
C) Aluminium wire  
D) Optical fibre✓

## Waves

- Q.1 An oscillating body is at mean position at  $t = 0$ . At  $t = T/4$  it will be at  
A) Extreme position✓  
C) Mean position  
B) Between extreme and mean position  
D) Beyond extreme position
- Q.2 In a simple pendulum, the tension of the string is  
A)  $g \cos \theta$   
B)  $mg \cos \theta$ ✓  
C)  $mg \sin \theta$   
D)  $mg$
- Q.3 Two sound waves having the same amplitudes are moving in the same direction out of phase. The amplitude of the resultant wave is  
A) Zero amplitude✓  
C) The sum of amplitude of the two waves  
B) Difference of the amplitudes of the two waves  
D) Double the amplitude of either wave
- Q.4 A source 'Y' of unknown frequency produces 4 beats with a source of 240 Hz and beats with a sound of 252 Hz. Frequency of the source 'Y' is  
A) 244 Hz✓  
B) 248 Hz  
C) 236 Hz  
D) 246 Hz
- Q.5 An organ pipe closed at one end has a length of 25 cm. Wavelength of its fundamental note is  
A) 25 cm  
B) 100 cm✓  
C) 50 cm  
D) 75 cm

The path difference 'BD' for destructive interference is

- A)  $(m + \frac{1}{2})\lambda$  ✓  
C)  $m\lambda$

- B)  $d \sin \theta$   
D)  $3\lambda$

Two waves of slightly different frequencies and traveling in same direction produce

- A) Interference.  
C) Polarization.

- B) Stationary waves.  
D) Beats. ✓

What is it that we use to calculate the speeds of distant stars and galaxies?

- A) Doppler Effect ✓  
C) Interference

- B) Beats  
D) All of the above

If the mass attached with a spring becomes four times, the time period of vibration becomes:

- A) One fourth  
C)  $3/4$

- B) Half  
D) Double ✓

Speed of the waves is equal to:

- A)  $f\lambda$

- B) Both A and B ✓

- C)  $\frac{\lambda}{T}$

- D)  $\lambda T$

In R-L Series circuit, the phase difference between applied voltage and current is given by the angle  $\theta$  which is:

A)  $\theta = \tan^{-1} \frac{LR}{\omega}$

B)  $\theta = \tan^{-1} \frac{\omega}{g}$  ✓

D)  $\theta = \tan^{-1} \frac{\omega R}{L}$

C)  $\theta = \tan^{-1} \omega LR$

A standing wave pattern is formed when the length of string is an integral multiple of \_\_\_\_\_ wavelength.

- A) Triple.  
C) Full.

- B) Half. ✓  
D) Double.

The angular frequency of simple pendulum is directly proportional to \_\_\_\_\_

- A)  $l$ .  
C)  $1/l$ . D)  $v/l$ . ✓

- B)  $v l$ .

Two waves of slightly different frequencies and travelling in the same direction lead to:

- A) Stationary Waves  
C) Interference

- B) Beats ✓  
D) Both B and C

In gases, the speed of sound is inversely proportional to \_\_\_\_\_ of the density when other factors are same.

- A) Square root. ✓  
C) Square.

- B) Third power.  
D) Third root.

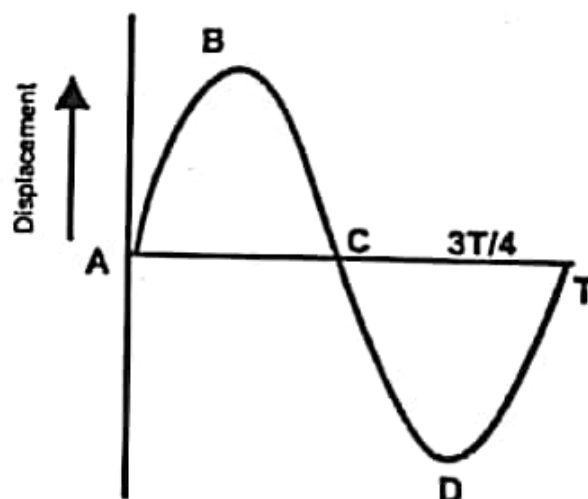
A 2m long pipe is open at both ends. What is its harmonic frequency?

- A) 42.5 Hz.  
C) 85 Hz. ✓

- B) 220 Hz.  
D) None of these.

At high temperature, the proportion of \_\_\_\_\_ wavelength radiation increase.

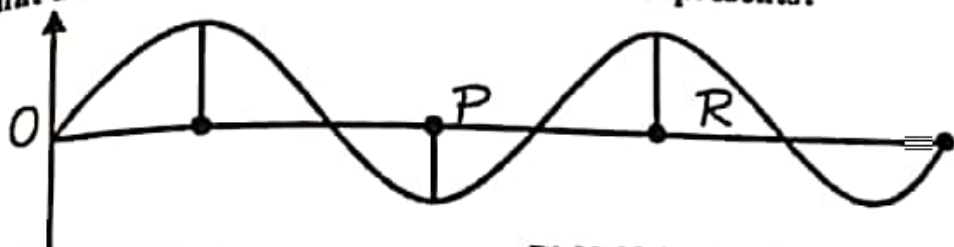
- A) AM radio.  
 C) Long radio.
- B) Shorter.  
 D) Both A and C.✓
- Q.18 Transverse waves cannot be setup in \_\_\_\_\_  
 A) Metals.  
 C) Solids.
- B) Fluids.✓  
 D) Soil.
- Q.19 A source of sound wave emits waves of frequency ' $f$ '. If ' $v$ ' is speed of sound waves, then what will be the wavelength of the waves
- A)  $\frac{v}{f}$  ✓  
 C)  $vf$
- B)  $\frac{v - u_o}{f}$   
 D)  $(v - u_o)f$
- Q.20 The spectrum of a star's light is measured and the wavelength of one of the lines is the sodium's line is found to be 589 nm. The same line has the wavelength of 497 nm when observed in the laboratory. This means the star is
- A) Moving away from the earth ✓  
 C) Moving towards the north
- B) Stationary  
 D) Revolving around the planet
- Q.21 What is the period of mass spring system during SHM if the ratio of mass to spring constant is  $\frac{1}{4}$ ?
- A)  $\pi$  ✓  
 C)  $2\pi$
- B)  $1/\pi$   
 D)  $\frac{1}{2}\pi$
- Q.22 Waveform of SHM is given in figure. At what time/times displacement is equal to zero?



- A)  $T/4$  only  
 C)  $3T/4$  only
- B)  $0, T/4, 3T/4$  and  $T$   
 D)  $0, T/2$  and  $T$  ✓
- Q.23 A simple harmonic oscillator has a time period of 10 seconds. Which equation relates its acceleration ' $a$ ' and displacement ' $x$ '?
- A)  $a = -2x$   
 C)  $a = -(20\pi)x$
- B)  $a = -\left(\frac{2\pi}{10}\right)^2 x$  ✓  
 D)  $a = -(20\pi)^2 x$
- Q.24 When the length of a simple pendulum is doubled, find the ratio of the new frequency to the old frequency?
- A)  $1/4$   
 C)  $1/2$
- B)  $\sqrt{2}$   
 D)  $1/\sqrt{2}$  ✓



Q.25 In the diagram below, the displacement of an oscillating particle is plotted against time. What does the length 'PR' on the time axis represents?



- A) Twice the frequency  
 C) Half the period ✓  
 B) Half the frequency  
 D) Twice the period
- Q.26 When the source of sound moves towards the stationary observer, the value of apparent frequency 'fo' is:

A)  $f_o = \left( \frac{v + u_i}{v} \right) f$   
 B)  $f_o = \left( \frac{v}{v + u_i} \right) f$   
 C)  $f_o = \left( \frac{v}{v - u_i} \right) f$  ✓  
 D)  $f_o = \left( \frac{v - u_i}{v} \right) f$

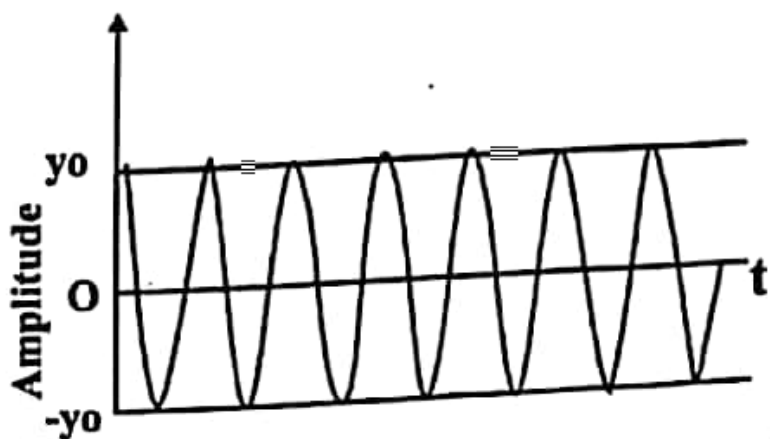
Q.27 For vibrating mass-spring system, the expression of kinetic energy at any displacement 'x' is given by:

A)  $\frac{1}{2} kx_o^2 \left( 1 - \frac{x^2}{x_o^2} \right)$   
 B)  $\frac{1}{2} m\omega \left( 1 - \frac{x^2}{x_o^2} \right)$   
 C)  $\frac{1}{2} kx_o^2$   
 D)  $\frac{1}{2} m\omega^2 x_o^2$  ✓

Q.28 Speed of sound through a gas is measured as 340 m/s at pressure P1 and temperature T1. What will be the speed of sound if pressure of gas is doubled but temperature is kept constant?

- A) 342 m/s ✓  
 B) 170 m/s  
 C) 340 m/s  
 D) 680 m/s

Q.29 Variation of amplitude with respect to time for an oscillation object is shown in figure.



Identify the oscillation:

- A) Damped  
 C) Critical  
 B) Undamped ✓  
 D) Heavily damped

Q.30 In a simple harmonic motion with a radius ' $x_0$ ', the velocity of the particle at any point is:

A)  $v = \omega \sqrt{x_0^2 - x^2}$

B)  $v = \omega \sqrt{x_0 - x}$

C)  $v = \omega(x^2 - x_0^2)$  ✓

D)  $v = \omega \sqrt{x - x_0}$

## Deformation of Solids

Q.1 Which of the following is the most ductile?

A) Glass.

B) Cast iron.

C) Copper. ✓

D) High carbon steel.

Q.2 A wire is stretched by a force which causes an extension. The energy is stored in it only when:

A) The extension of wire is proportional to force applied ✓

C) The cross-section area of the wire remains constant

B) The wire is not stretched beyond its elastic limit

D) The weight of wire is negligible

Q.3 Which statement is correct:

A) Elasticity is that property of body which enables body to regain its original dimension

C) Elasticity is that property of a body that does not allow it to return to its original shape

B) Elasticity is that property of a body that allows it to retain its original shape and dimension after the stress ✓  
is removed.

D) Elasticity is that property of a body that obeys Hooke's law.

Q.4 The ratio of tensile strength to tensile strain is called

A) Modulus of elasticity

B) Young's Modulus ✓

C) Bulk Modulus

D) Shear Modulus

Q.5 A wire is stretched by a force ' $F$ ' which causes an extension  $\Delta l$ , the energy stored in the wire is:

A)  $F\Delta l$

B)  $\frac{1}{2} F\Delta l^2$

C)  $2F\Delta l$

D)  $\frac{1}{2} F\Delta l$  ✓

Q.6 The stress-strain graph, deduced the following limits successively:

A) Proportional limit, yield limit, elastic limit

C) Yield limit, elastic limit, proportional limit

B) Proportional limit, elastic limit, yield limit ✓

D) Elastic limit, proportional limit, yield limit

Q.7 A 4.0 m long wire is subjected to stretching force and its length increases by 40 cm. The percent elongation which the wire undergoes is:

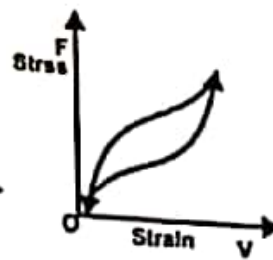
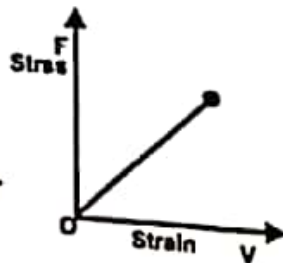
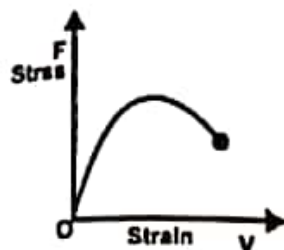
A) 0.10 %

B) 10 % ✓

C) 40 %

D) 20 %

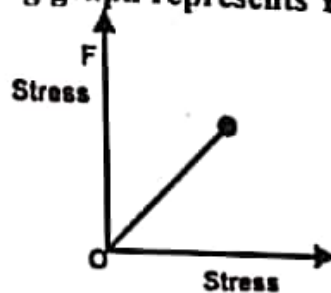
Q.8 Three graphs for three types of materials are shown in the figure.



Which row describes the correct materials?

- |    | X       | Y       | Z        |
|----|---------|---------|----------|
| A) | Brittle | Ductile | Polymer  |
| B) | Brittle | Polymer | Ductile  |
| C) | Polymer | Brittle | Ductile  |
| D) | Ductile | Brittle | Polymer✓ |

Q.9 Which feature of the following graph represents Young's Modulus?



- A) Area under graph  
 C) Gradient of the graph  
 B) Reciprocal of the gradient✓  
 D) Product of gradient and area of the curve.

Q.10 Strain energy in a deformed energy is stored in the form of:

- A) Elastic Energy  
 B) Plastic Energy  
 C) Potential Energy✓  
 D) Kinetic Energy

Q.11 A wire of area of cross section 'A' and original length 'l' is subjected to a load 'L'. A second wire of same material with an area is '2A' and length '2l' is subjected to the same load 'L'. If the extension in first wire is 'X' and second wire is 'Y', find the ratio 'X/Y'.

- A)  $\frac{1}{4}$   
 B)  $\frac{1}{1}$ ✓  
 C)  $\frac{1}{2}$   
 D)  $\frac{1}{1}$

Q.12 The ratio of applied stress to the volumetric strain is called:

- A) Bulk Modulus✓  
 B) Tensile modulus  
 C) Shear Modulus  
 D) Young's Modulus

Q.13 The wire made of copper belong to which specific kind of material:

- A) Ductile material✓  
 B) Brittle material  
 C) Tough material  
 D) Deformed material

Q.14 A 1.25cm diameter cylinder is subjected to load of 2500kg, stress on bar is:

- A) 200 Pa  
 B)  $2 \times 10^6$  Pa  
 C)  $2 \times 10^5$  Pa✓  
 D)  $2 \times 10^9$  Pa



- Q.15 A wire of length 2m is attached with mass of 5kg vertically, tensile strain of wire is  $0.3 \times 10^{-3}$ , the extension in wire is:  
 A) 1.5 mm B) 0.5 mm  
 C) 2mm D) 0.6 mm ✓
- Q.16 In the case of linear deformation, the ratio of tensile stress to tensile strain is called:  
 A) young's double slit phenomenon B) Young's modulus ✓  
 C) Bulk modular D) energy stored in stretched wire
- Q.17 Which material will follow the below stress strain curve:  
 DIA  
 A) Iron B) Lead  
 C) Glass ✓ D) Copper
- Q.18 The sum of all forms of molecular energies (Kinetic and potential) of a substance is termed as?  
 A) Elastic energy B) Internal energy ✓  
 C) Absolute energy D) Heat energy
- Q.19 The area under the extension-load graph of an elastic material whose elastic limit has not been exceeded gives its:  
 A) Stress B) Strain  
 C) Strain energy ✓ D) Young modulus
- Q.20 A wire has a spring constant of  $5 \times 10^4 \text{ N m}^{-1}$ . It is stretched by a force to extension of 1.4 mm. Calculate the strain energy stored in the wire.  
 A)  $4.9 \times 10^{-5} \text{ J}$  B) 4.9 J  
 C)  $4.9 \times 10^{-2} \text{ J}$  ✓ D)  $4.9 \times 10^{-3} \text{ J}$

## Ideal Gases

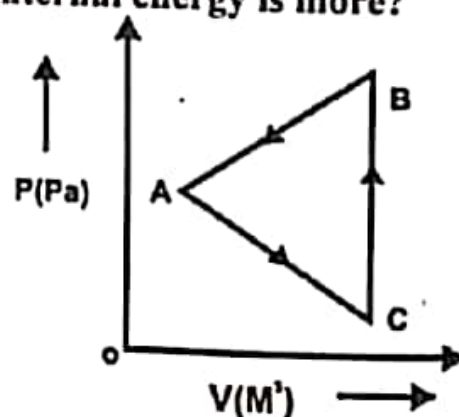
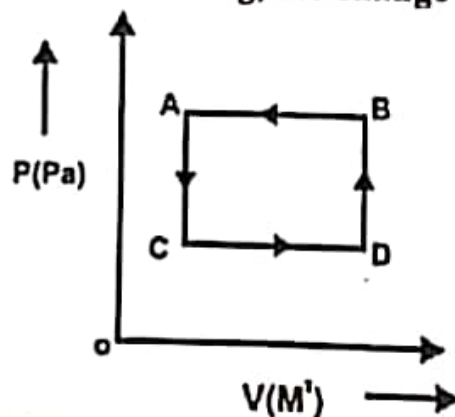
- Q.1 The value of universal; Gas Constant 'R' is:  
 A)  $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$  B)  $1.38 \text{ J mol}^{-1} \text{ K}^{-1}$   
 C)  $1.38 \text{ J mol}^{-1} \text{ K}^{-2}$  D)  $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$  ✓
- Q.2 Which one of the following compounds show cis-trans isomerism?  
 A) 1-butene B) 1-bromo-2-chloropropane ✓  
 C) 1-hexene D) Propene
- Q.3 Which of the following is the expression of root mean square speed of a gas having number of molecules contained in the container?  
 A)  $\sqrt{\frac{v_1^2 + v_2^2 + \dots + v_x^2}{N}}$  ✓ B)  $\sqrt{\frac{v_1 + v_2 + \dots + v_x}{N}}$   
 C)  $\frac{v_1^2 + v_2^2 + \dots + v_x^2}{N}$  D)  $\frac{v_1 + v_2 + \dots + v_x}{N}$
- Q.4 For a gas of volume V in its equilibrium state, if the pressure does change with time then total kinetic energy of gas is constant because  
 A) Collisions between gas molecules occur

- Q.5 C) Collisions between gas molecules occur linearly  
 B) Collisions must be elastic ✓  
 H<sub>2</sub> and O<sub>2</sub> both are at thermal equilibrium at temperature 300 K. Oxygen molecule is 16 times massive than hydrogen. Root mean square speed of hydrogen is  
 A) 4 root mean square of oxygen ✓  
 C) 1/4 root mean square of oxygen  
 B) 1/16 root mean square of oxygen  
 D) 1/6 root mean square of oxygen
- Q.6 Which of the following is expression of mean square speed of 'N' gas molecules contained in a cylinder?  
 A)  $\frac{v_1 + v_2 + \dots + v_x}{N}$  ✓  
 C)  $\frac{v_1^2 + v_2^2 + \dots + v_x^2}{N}$   
 B)  $\sqrt{\frac{v_1 + v_2 + \dots + v_x}{N}}$   
 D)  $\sqrt{\frac{v_1^2 + v_2^2 + \dots + v_x^2}{N}}$
- Q.7 What is the value of universal gas constant?  
 A) 8314 Jmol<sup>-1</sup>K<sup>-1</sup> ✓  
 C) 831.4 Jmol<sup>-1</sup>K<sup>-1</sup>  
 B) 83.14 Jmol<sup>-1</sup>K<sup>-2</sup> ✓  
 D) 8.314 Jmol<sup>-1</sup>K<sup>-1</sup>
- Q.8 A gas sample contains three molecules each having speed 1 ms<sup>-1</sup>, 2 ms<sup>-1</sup>, 3 ms<sup>-1</sup>. What is the mean square speed?  
 A) 14/3 m/s ✓  
 B) 2 m/s  
 C) 6 m/s  
 D)  $\sqrt{14/3}$  m/s
- Q.9 A gas containing 'N' number of molecules of a gas having mass of each molecule 'm' is in a cubic container having length of each side 'a'. What is the density of gas contained in cube?  
 A)  $N/a^2$   
 C)  $m/a^3$   
 B)  $Nm/a^3$  ✓  
 D)  $Na^3/m$
- Q.10 In 'General Gas Equation PV=nRT', 'n' represents the number of moles of gas. Which of the following represents the relation of 'n'?  
 A)  $n = NN_A$   
 C)  $n = N/N_A$  ✓  
 B)  $n = N_A/N$   
 D)  $n = N + N_A$
- Q.11 Two sample of gases '1' and '2' are taken at same temperature and pressure but the ratio of number of their volume is V<sub>1</sub>:V<sub>2</sub> = 2:3. What is the ration of number of moles of the gas sample?  
 A) 3:2  
 C) 4:9  
 B)  $\sqrt{2} : \sqrt{3}$   
 D) 2:3 ✓
- Q.12 Root mean square velocity of a gas having pressure 'P' and density 'ρ' is given by:  
 A)  $\sqrt{\frac{3P}{\rho}}$  ✓  
 C)  $\sqrt{\frac{3\rho}{P}}$   
 B)  $\frac{3P}{\rho}$   
 D)  $\frac{3\rho}{P}$
- Q.13 The relation  $\frac{R}{N_A} = 1.38 \times 10^{-23} \text{ JK}^{-1}$  in a gas law is known as:  
 A) Avogadro's Constant  
 C) Charles constant  
 B) Newton's constant  
 D) Boltzmann's constant ✓
- Q.14 The relation 'PV = nRT' shows which law of physics:  
 A) Charles Law  
 C) Avogadro's Law  
 B) Newton's Constant  
 D) Ideal Gas Law ✓



# Heat and Thermodynamics

- Q.1 The value of Wien's constant is  
 A)  $2.90 \times 10^{-3} \text{ mK}$  ✓  
 B)  $4.22 \times 10^{-7} \text{ mK}$   
 C)  $3.34 \times 10^{-4} \text{ mK}$   
 D)  $3.42 \times 10^{-8} \text{ mK}$
- Q.2 The turbine in a steam power plant takes steam from a boiler at  $427^\circ\text{C}$  and exhausts into a low temperature reservoir at  $77^\circ\text{C}$ . What is the maximum possible efficiency?  
 A) 50% ✓  
 B) 60%  
 C) 40%  
 D) 70%
- Q.3 Which one is not an irreversible process?  
 A) Slow compression of a gas into a cylinder  
 B) Explosion  
 C) Changes due to friction ✓  
 D) Dissipation of energy
- Q.4 The value of Stefan's Boltzmann Constant is:  
 A)  $4.28 \times 10^{-7} \text{ Wm}^{-2}\text{K}^{-4}$   
 B)  $3.62 \times 10^{-4} \text{ Wm}^{-2}\text{K}^{-4}$   
 C)  $4.28 \times 10^{-4} \text{ Wm}^{-2}\text{K}^{-4}$   
 D)  $5.67 \times 10^{-5} \text{ Wm}^{-2}\text{K}^{-4}$  ✓
- Q.5 The pressure on the other sides and everywhere inside the vessel will be according to the:  
 A) Pascal's Law ✓  
 B) Boyle's Law  
 C) Hook's Law  
 D) Charles's Law
- Q.6 The value of universal gas constant is 17  
 A)  $8.314 \text{ Jmol}^{-1}\text{K}^{-1}$  ✓  
 B)  $7.23 \text{ Jmol}^{-1}\text{K}^{-1}$   
 C)  $8.324 \text{ Jmol}^{-1}\text{K}^{-1}$   
 D)  $1.00 \text{ Jmol}^{-1}\text{K}^{-1}$
- Q.7 For adiabatic process, the First Law of Thermodynamics is:  
 A)  $W = \Delta U + Q$   
 B)  $Q = W$   
 C)  $Q = -W$   
 D)  $W = -\Delta U$  ✓
- Q.8 The entropy of the universe always:  
 A) Decreases  
 B) Remains the same  
 C) Increases ✓  
 D) Both A and B
- Q.9 In which of the following, the change in internal energy is more?

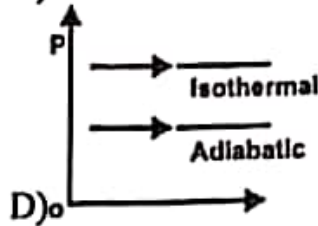
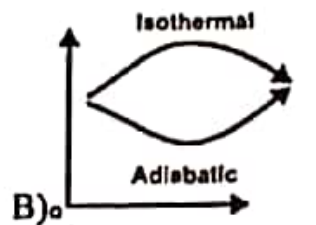
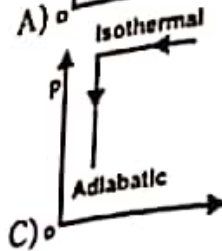
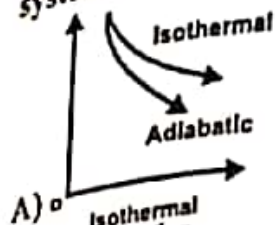


- A) In system A  
 B) In system B  
 C) Cannot be predicted  
 D) Change is zero in both. (both are cycles) ✓

- Q.10 Pressure volume graph of two systems 'A' and 'B' are plotted under isothermal and

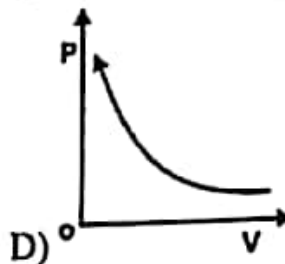
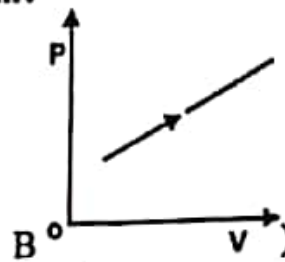
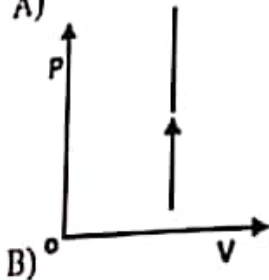
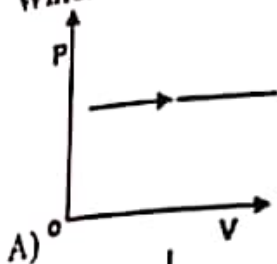


Dogar's Unit 1  
adiabatic conditions. Which of the following observation of graph represents the two systems?



Answer: A

Q.11 Which of the following curve is an isotherm?



Answer: D

Q.12 If 'Q' is the amount of heat supplied to a system and 'W' is the work done, then change in internal energy can be defined as

A)  $Q/W$

B)  $W/Q$

C)  $Q - W$  ✓

D)  $1 + Q/W$

Q.13 A heat engine operating according to second law of thermodynamics rejects one fourth of the heat taken from high temperature reservoir. What is the percentage efficiency of heat engine?

A) 100%

B) 50%

C) 25%

D) 75% ✓

Q.14 First law of thermodynamics under adiabatic conditions can be mathematically written as:

A)  $Q = W$

B)  $Q = U + W$

C)  $Q = \Delta U$

D)  $W = -\Delta U$  ✓

Q.15 What is the factor upon which change in internal energy of an ideal gas depends?

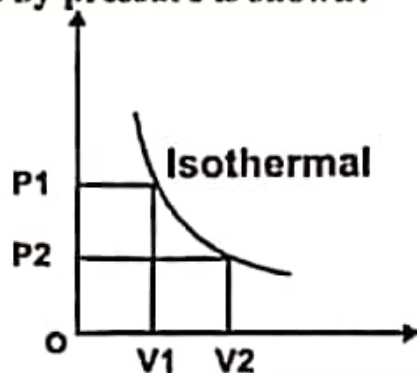
A) Change in volume

C) Change in temperature and volume

B) Change in temperature ✓

D) Path followed to change internal energy

- Q.16 What will be the mathematical form of first law of thermodynamics for a system whose variation of volume by pressure is shown?



- A)  $Q = U$   
 C)  $U = W$   
 B)  $Q = U/W$   
 D)  $Q = W$ ✓

- Q.17 For a heat engine 'A' ratio of  $Q_1$  to  $Q_2$  is  $2/3$  while that of heat engine 'B', ratio of  $Q_2$  to  $Q_1$  is  $1/3$ . What is the value  $\eta_A : \eta_B$ ?

- A) 1:3  
 C) 1:2✓  
 B) 2:3  
 D) 2:1

- Q.18 At triple point of water, the pressure of gas is 2680 Pa, by changing 'T' the pressure increases to 4870 Pa. Then 'T' is:

- A) 496.38 K✓  
 C) 438.96 K  
 B) Zero  
 D) 496.38 °F

## Electronics

- Q.1 The grid in the cathode ray oscilloscope \_\_\_\_\_.

- A) Controls number of waves  
 C) Controls the brightness of spot formed✓  
 B) Accelerates electrons  
 D) Has positive potential with respect to cathode

- Q.2 In transistors, the base region is very thin, of the order of

- A)  $10^{-5}$  cm  
 C)  $10^{-6}$  m✓  
 B)  $10^{-6}$  mm  
 D)  $10^{-6}$   $\mu$ m

- Q.3 The closed loop gain of OP-AMP depends on

- A) Internal structure of OP-AMP  
 C) Externally connected resistances✓  
 B) Voltage of power supplies  
 D) Input resistance

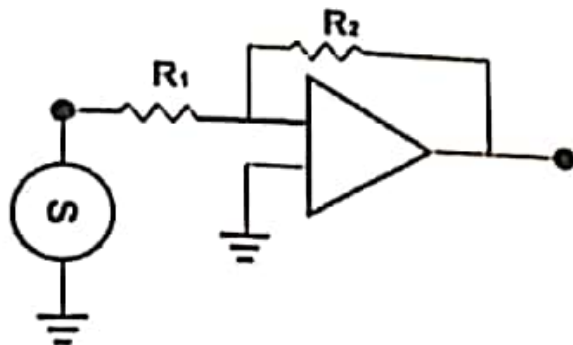
- Q.4 The net charge on an N-type substance is

- A) 0.7 volts✓  
 C) 0.3 volts  
 B) 0.25 volts  
 D) 0.07 volts

- Q.5 The substances like germanium and silicon have

- A) Negative temperature coefficients✓  
 C) Positive temperature coefficients  
 B) Both A and B  
 D) None of the above

- Q.6 If  $R_1 = 10 \text{ k}\Omega$  and  $R_2 = 100 \text{ k}\Omega$  then the gain of op-amplifier as inverting amplifier is:



A) -1

C) 10

If inputs  $A = 1$ ,  $B = 0$  and output  $X = 1$ , then it corresponds to the operation of a:

A) AND Gate

C) NAND Gate✓

B) -10✓

D) 1

B) XNOR Gate

D) NOR Gate

In a certain circuit, if the transistor has a collector current of 10 mA and base current of 50  $\mu$ A, then the current gain of the transistor is:

A) 250

C) 100

B) 150

D) 200✓

A signal that is applied at the inverting input terminal of an op-amplifier undergo amplification, at the output terminal with a phase shift of:

A)  $0^\circ$

C)  $270^\circ$

B)  $360^\circ$

D)  $180^\circ$ ✓

To convert the Si crystal into p-type semi-conductor, which group element will be doped:

A) Trivalent Element✓

C) Second Group Element

B) Fourth Group Element

D) Pentavalent Element

In which of the following, output is similar to NAND gate if input  $A=0$  and input  $B=1$ .

A) NOR.

C) XNOR.

B) XOR.✓

D) Both B and C.

In LED when an electron combines with a \_\_\_\_\_ during forward bias conduction, a photon of visible light is emitted.

A) High voltage.

C) Photon.

B) Hole.✓

D) Positron.

Practically \_\_\_\_\_ current flows in a reverse biased p-n junction.

A) No.

C) Very large.

B) Few milliamperes.

D) Both A and C.✓

Cesium coated oxidized silver emits electrons for \_\_\_\_\_ light.

A) Infrared.✓

C) Ultraviolet.

B) Visible.

D) Green.

Which device is the most efficient?

A) Nuclear reactor.✓

C) Storage battery.

B) Silicon solar cell.

D) Dry battery cell.

An n-type semi-conductor is made by doping silicon crystal with \_\_\_\_\_

A) Indium.

C) Aluminium.

B) Arsenic.✓

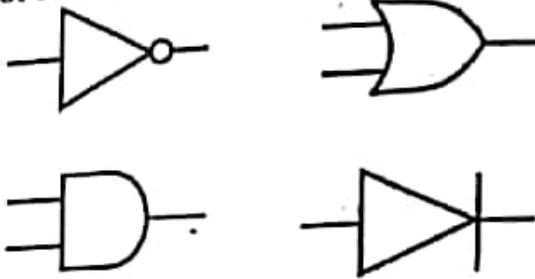
D) Both B and C.

Which of the following is the proper way to study the sinusoidal waveform of the

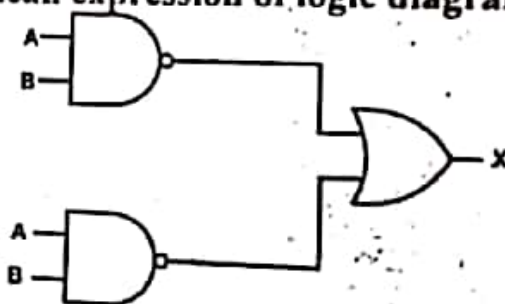


voltage?

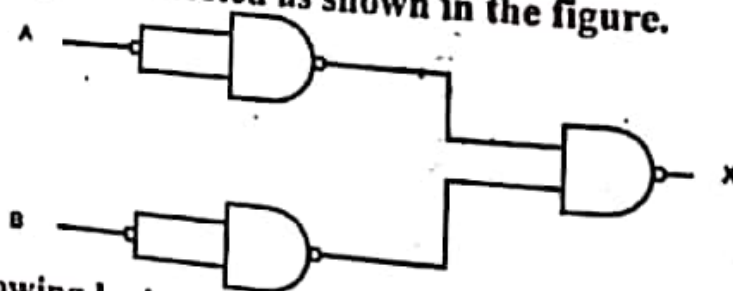
- A) Voltage is connected to X input and the time base is switched off  
 C) Voltage is connected to Y input and the time base is switched on ✓  
 B) Voltage is connected to Y input and the time base is switched off  
 D) Voltage is connected to X input and the time base is switched on
- Q.18 Electron gun in cathode ray oscilloscope contains  
 A) Filament, cathode, grid, anodes ✓  
 C) Cathode, anode, capacitor, screen  
 B) Emitter, base, collector  
 D) Resistance, capacitor, inductor
- Q.19 What is the logic symbol for a NOT Gate?



- Q.20 The voltage that is applied across X-plates is provided by a circuit called  
 A) Audio generator  
 C) Time base generator ✓  
 B) Signal generator  
 D) Linear generator
- Q.21 Which of the following is the proper way to study the sinusoidal wave form voltage?  
 A) Voltage is connected to 'Y' input and time base is switched on. ✓  
 C) Voltage is connected to 'X' input and time base is switched off.  
 B) Voltage is connected to 'Y' input and time base is switched off.  
 D) Voltage is connected to 'X' input and time base is switched on.
- Q.22 What is the output Boolean expression of logic diagram shown in figure below:



- A)  $\overline{(A+B)} \cdot \overline{(A+B)}$   
 B)  $\overline{A} \cdot \overline{B} + \overline{A} \cdot \overline{B}$   
 C)  $(\overline{A} + \overline{B})(\overline{A} + \overline{B})$   
 D)  $\overline{A} \overline{B} + \overline{A} \overline{B}$  ✓
- Q.23 Three NAND gates are connected as shown in the figure.



- Which of the following logic gate is formed in the connected circuit?  
 A) OR ✓  
 C) AND  
 B) NOR  
 D) NAND
- Q.24 What is the output of the truth table?

Output  $x = AB + \overline{A}B$

| A | B |
|---|---|
| 0 | 0 |
| 1 | 0 |
| 1 | 1 |

A)

| X |
|---|
| 0 |
| 0 |
| 1 |
| 1 |

✓

B)

| X |
|---|
| 1 |
| 1 |
| 1 |
| 0 |

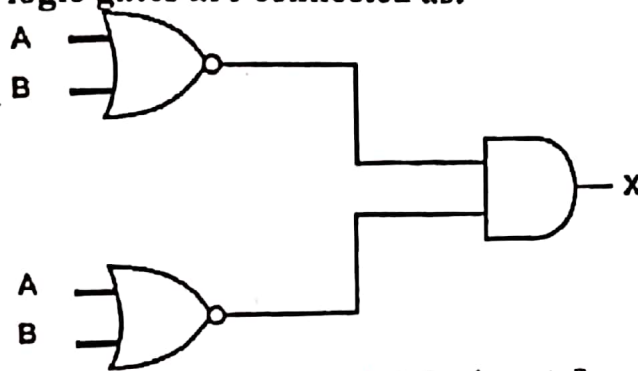
C)

| X |
|---|
| 1 |
| 0 |
| 0 |
| 1 |

D)

| X |
|---|
| 0 |
| 1 |
| 1 |
| 1 |

If the fundamental logic gates are connected as:



What are the mathematical notation for this logic gate?

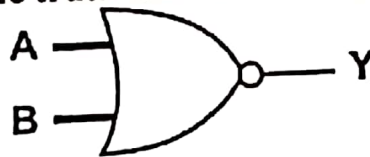
A)  $(A+B).(A+B)$

B)  $\overline{(A+B)}(A+B)$

C)  $\overline{(A+B)}.\overline{(A+B)}$  ✓

D)  $\overline{AB} + \overline{AB}$

Which of the following is the truth table for the logic gate;



A)

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

✓

B)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

C)

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

D)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

- Q.27 Which one of the following is the Boolean expression of NAND gate?  
 A)  $X = A.B$   
 B)  $X = A.B$  ✓  
 C)  $X = A + B$   
 D)  $X = A + B$

- Q.28 Which one of the following is the truth table of NAND gate?

A)

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

B)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

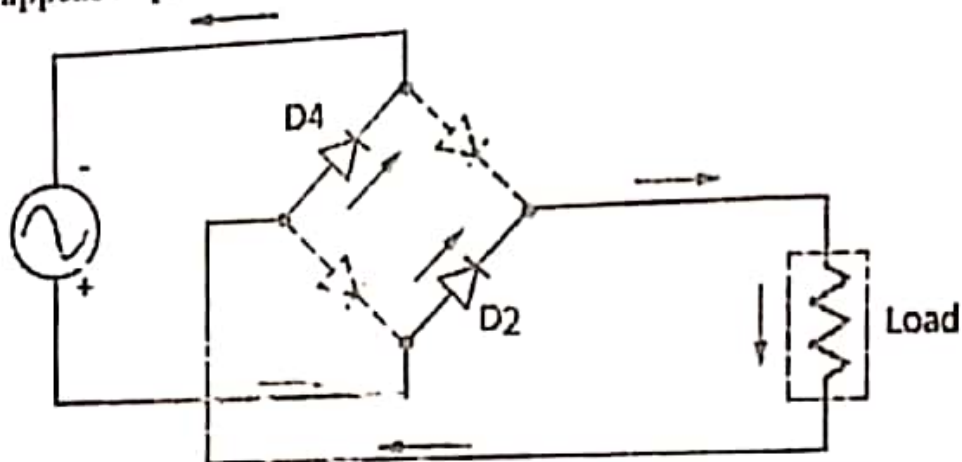
C)

| A | B | Y |
|---|---|---|
| 0 | 1 | 1 |
| 1 | 1 | 0 |

D)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 1 | 1 |

- Q.29 What happens in positive cycle of AC input?



- A)  $D_1$  and  $D_3$  conducts  
 B)  $D_3$  and  $D_4$  conducts  
 C)  $D_1$  and  $D_2$  conducts  
 D)  $D_2$  and  $D_4$  conducts ✓

- Q.30 If signal is applied to input of non-inverting amplifier through resistance of  $1\text{ k}\Omega$ , and the value of feedback resistance is  $10\text{ k}\Omega$ , the gain is:  
 A) 11  
 B) 1.1 ✓  
 C) 10  
 D) 0.11

## Current Electricity

- Q.1 The consumption of energy by 60-watt bulb in 2 seconds is:  
 A) 20 J  
 B) 30 J  
 C) 120 J ✓  
 D) 0.02 J
- Q.2 Electric intensity is a vector quantity and its direction is  
 A) Perpendicular to the direction of field  
 B) At a certain angle  
 C) Opposite to the direction of force  
 D) Along the direction of force ✓
- Q.3 The magnitude of an electric field between two separated plates can be calculated the relation



A)  $\Delta V = Ed$ ✓

B)  $\Delta V = \frac{E}{q_0}$

C)  $\Delta V = E/d$

D)  $E = \frac{d}{\Delta V}$

SI unit of electric flux is

A)  $NmC^{-1}$

B)  $Nm^2C^{-2}$ ✓

C)  $Nm^{-2}C^{-2}$

D)  $Nm^2C^{-2}$

The equivalent current which passes from a point at higher potential to a point at a lower potential as if it represented a movement of positive charges is

A) Electronic current

B) Magnetic lines

C) Electric current

D) Conventional current✓

If 'V' is applied potential difference across a resistance 'R', then loss in potential energy per unit time is

A)  $VI$

B)  $\frac{V^2}{R}$

C)  $I^2R$

D) All of the above✓

The sensitivity of a galvanometer can be decreased by

A) Increasing magnetic field

B) Increasing  $\frac{c}{BAN}$  Ratio✓

C) Increasing number of turns of the coil

D) Decreasing length of couple 'c'

The work done in moving a unit positive charge from one point to another against the electric field is a measure of:

A) Capacitance

C) Potential difference between two points✓

B) Intensity of electric field

D) Resistance between two points

The deviation of I-V graph from the straight line is due to:

A) Decrease in temperature and decrease in resistance

C) Increase in temperature and increase in resistance✓

B) Decrease in temperature and increase in resistance

D) Increase in temperature and decrease in resistance

The fractional change in resistance per Kelvin is known as:

A) Temperature coefficient of resistance✓

C) Thermal coefficient

B) Linear coefficient of expansion

D) Volumetric coefficient of expansion

The energy supplied by the cell to the charge carriers is derived from the conversion of:

A) Heat energy into Electrical energy

C) Chemical energy into Electrical energy✓

B) Solar energy into Electrical energy

D) Mechanical energy into Electrical energy

Q.12 What shunt resistance must be connected across a Galvanometer of  $20\ \Omega$  resistance which gives full scale deflection with  $2.0\ \text{A}$  current, so as to convert it into an Ammeter of range  $10\ \text{A}$ ?

A)  $5\ \Omega$ ✓

B)  $3\ \Omega$

C)  $2\ \Omega$

D)  $4\ \Omega$

Q.13 The current measuring part of the Avometer consists of number of low resistances connected:

A) At an angle of  $180^\circ$  with the galvanometer

C) Parallel with the galvanometer✓

B) At an angle of  $45^\circ$  with the galvanometer

D) Perpendicular to the galvanometer

Q.14 Resistance in RC circuit of time constant 2 seconds is  $1000\ \text{Ohms}$ . What is value of C in the circuit?

A)  $2\ \mu\text{farad}$ .

B)  $200\ \mu\text{farad}$ .

C)  $20\ \mu\text{farad}$ .✓

D)  $2000\ \mu\text{farad}$

Q.15 The algebraic sum of potential changes in a closed circuit is zero is Kirchhoff's \_\_\_\_\_ rule.

A) First.

B) Third.

C) Second.✓

D) None of these.

Q.16 A  $5\ \text{Ohm}$  resistor is indicated by a single \_\_\_\_\_ color band around its body.

A) Red.

B) Blue.

C) Green.✓

D) Brown.

Q.17 A wire has resistance  $100\ \text{Ohm}$  at  $0^\circ\text{C}$  and  $200\ \text{Ohm}$  at  $100^\circ\text{C}$ . What is its temperature coefficient in  $\text{K}^{-1}$ ?

A)  $-0.01$ .

B)  $0.01$ .

C)  $-1/273$ .

D)  $1/273$ .✓

Q.18 Which of the following has the highest resistivity?

A) Germanium.

B) Copper.

C) Silver. ✓

D) Platinum.

Q.19 If inductance is denoted by  $L$  and resistance by  $R$ , which of the following is true for a choke?

A)  $R$  is large,  $L$  is very small.

B) Both  $R$  and  $L$  are large.

C)  $R$  is very small,  $L$  is large.✓

D) Both  $R$  and  $L$  are very small.

Q.20 The heat produced by a current  $I$  in the wire of resistance  $R$  during time interval  $t$  is

A)  $I^2/Rt$ .

B)  $I^2/Rt$ .

C)  $I^2Rt$ .✓

D)  $IR^2t$ .

Q.21 If  $2\ \text{A}$  current passes through a resistor when connected to a certain battery. If the resistance is replaced by the double resistance, then the current will become

A)  $2\ \text{A}$

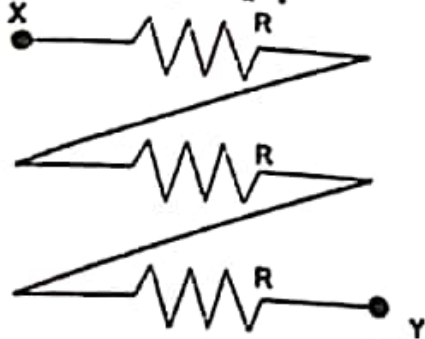
B)  $6\ \text{A}$

C)  $4\ \text{A}$

D)  $1\ \text{A}$ ✓

Q.22 Three resistors each having value ' $R$ ' are connected as shown in figure. What is the

equivalence resistance between 'X' and 'Y'?



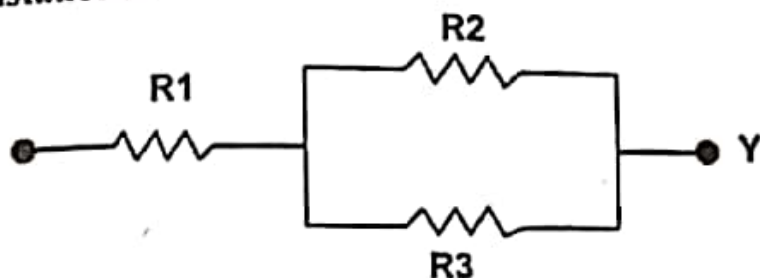
A)  $3R$  ✓

C)  $R$

Three resistors of resistance  $R_1$ ,  $R_2$  and  $R_3$  are connected as shown in figure. Equivalence resistance is:

B)  $R/3$

D)  $R^3$



A)  $R_1 + R_2 + R_3$

B)  $\frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1 + R_2}$  ✓

B)  $\frac{R_1 + R_2 + R_3}{R_1 R_2}$

D)  $\frac{R_1 R_2 R_3}{R_1 R_2}$

What will be the effect on the capacitance of a capacitor if area of each plate is doubled while separation between the plates is halved?

A) Capacitance remains same

B) Capacitance becomes four times ✓

C) Capacitance becomes double

D) Capacitance reduces to half

10 V potential difference is applied across the plate of  $1 \mu\text{F}$  capacitor. What is the energy stored in capacitor?

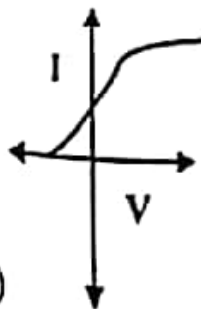
A) 0.5 mJ

B) 5 mJ

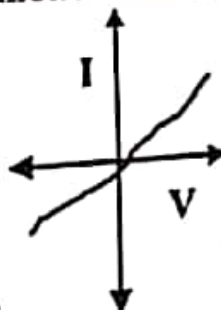
C) 0.05 mJ ✓

D) 50 mJ

Which one of the following is I-V curve of a junction diode?



A)



C)



Force experienced by a moving charge in a magnetic field is:

- A)  $F = \mu n l$
- C)  $F = BA \cos \theta$
- C)  $F = \mu_0 NI$

The value of permeability of free space  $\mu_0$  is:

- A)  $4\pi \times 10^{-7} \text{ WbA}^{-1}\text{m}^{-1}$  ✓
- C)  $4\pi \times 10^2 \text{ WbA}^{-2}\text{m}^{-2}$

A charge of two micro coulombs ( $2\mu\text{C}$ ) moves with velocity of two meter per second ( $2 \text{ m/sec}$ ) in the direction of two Tesla magnetic field. The force that will act on it will be:

- A)  $2 \text{ N}$
- C) Zero ✓

- B)  $8 \text{ N}$
- D)  $4 \text{ N}$

We have two coils placed close to each other. When we switch on the battery connected to primary coil while keeping the sliding contact of rheostat at fixed position, the reading of Galvanometer:

- A) First increases and then becomes zero ✓
- C) First increases and then becomes constant at some value
- B) Increases with the passage of time
- D) Remains zero

Power losses in a transformer can be minimized:

- A) By increasing turn ratio
- C) By decreasing turn ratio
- B) By stopping the flow of Eddy currents ✓
- D) Using material of the core whose hysteresis area is large

Frequency of L-C circuit will resonate under the driving action of the antenna by angular value of:

- A) Capacitance ✓
- C) Impedance
- B) Inductance
- D) Resistance

The Lenz's law refers to induced \_\_\_\_\_

- A) emf.
- C) Resistance.
- B) Shear.
- D) Currents. ✓

Tuning of the radio is the best example of electrical \_\_\_\_\_

- A) Resonance. ✓
- C) Resistance.
- B) Current.
- D) None of these.

In a step-down transformer the output current \_\_\_\_\_

- A) Is reduced.
- C) Is increased. ✓
- B) Remains same.
- D) None of these.

The net magnetic field created by the electrons within an atom is due to the field created by their \_\_\_\_\_ motion.

- A) Orbital.
- B) Orbital & spin.

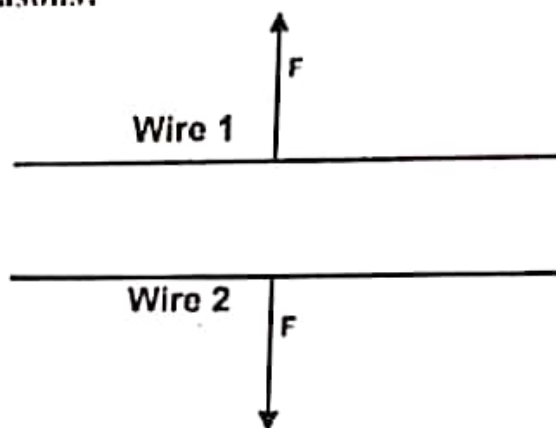
C) Spin.

D) Orbital  $\times$  spin. ✓

Q.13 If the number of turns of a solenoid circular coil is doubled, but the current in the coil and radius of the coil remains same, then what will be the magnetic flux density produced by the coil?

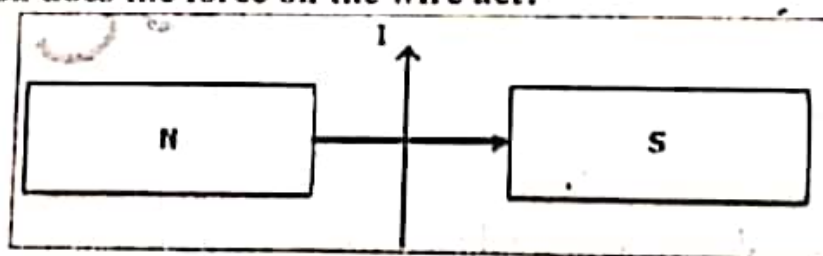
- A) Magnetic flux density will be halved
- C) Magnetic flux density increases by different amount at different points
- B) Magnetic flux density remains unchanged
- D) Magnetic flux density will be doubled ✓

Q.14 Two long parallel wires Wire 1 and Wire 2 repel each other as shown in the figure. What could be the reasons?



- A) Both carry current in same direction
- C) Both carry current in opposite direction ✓
- B) Wire 1 has current, but Wire 2 has no current
- D) Wire 2 has current, Wire 1 has no current

Q.15 The diagram shows a wire, carrying a current 'I', placed between the poles of a magnet. In which direction does the force on the wire act?

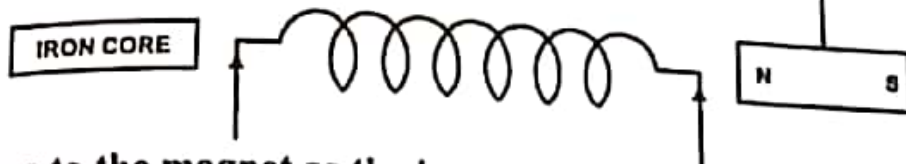


- A) Upwards
- C) Downwards ✓
- B) Towards the 'N' pole of the magnet
- D) Towards the 'S' pole of the magnet

Q.16 A 10 cm long solenoid has 100 turns. What will be the magnetic field inside it along its axis if one micro ampere current is passed through it?

- A)  $4\pi \times 10^{-13}$  tesla
- B)  $4\pi \times 10^{-10}$  tesla ✓
- C)  $4\pi \times 10^{-7}$  tesla
- D)  $4\pi \times 10^{-16}$  tesla

Q.17 The diagram shows a small magnet hanging on a thread near the end of a solenoid carrying a steady current 'I':

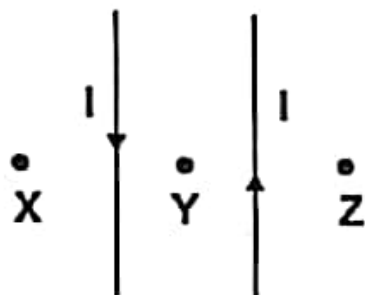


What happens to the magnet as the iron core is inserted into the solenoid?

- A) It moves towards solenoid and rotates through  $180^\circ$
- C) It moves towards the solenoid ✓
- B) It moves away from solenoid
- D) It moves away from solenoid and rotates through  $180^\circ$

Q.18

Two long straight parallel wires held vertically have equal but opposite currents as shown in the figure.



Which of the following effect will be observed?

- A) Magnetic field at 'X' is stronger than that at 'Y' and 'Z' ✓
- C) Magnetic field at 'X' is weaker than that at 'Y' and 'Z'
- B) Magnetic field at 'X', 'Y' and 'Z' is same
- D) Magnetic field at 'X' is weaker than that at 'Y' but stronger than that at 'Z'.

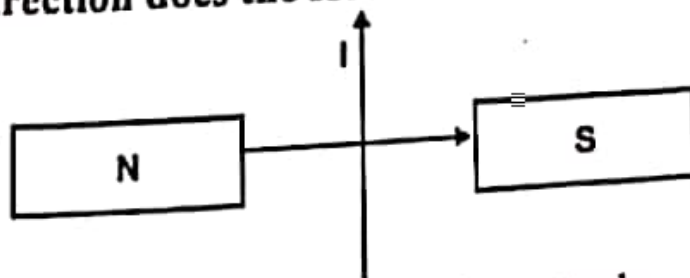
19 A solenoid is cut into two halves. Magnetic induction due to same current in each half will be:

- A) Half of the original
- B) Same as original ✓
- C) Double of the original
- D) Four times of the original

20 A long straight current carrying conductor has current directed from bottom to top when held vertically. What will be the direction of magnetic field lines when observed from below the conductor?

- A) Clockwise ✓
- B) Vertically upward
- C) Anti clockwise
- D) Vertically downward

21 The diagram shows a wire, carrying a current 'I', placed between the poles of magnet: In which direction does the force on the wire act?

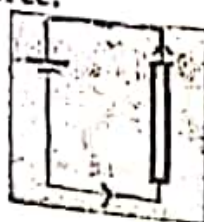


- A) Towards the 'N' pole of the magnet
- B) Upwards
- C) Downwards ✓
- D) Towards the 'S' pole of the magnet

2 A solenoid 15 cm long has 300 turns of wire. A current of 5 A flows through it. What



- is the magnitude of magnetic field inside the solenoid?  
 A)  $75 \times 10^7 \text{ T}$   
 B)  $4\pi \times 10^{-3} \text{ T}$  ✓  
 C)  $60 \times 10^{-3} \text{ T}$   
 D)  $750\pi \times 10^{-3} \text{ T}$
- Q.23 Due to current in a straight conductor the difference between magnetic field lines  
 A) Increases away from conductor ✓  
 B) Increases towards conductor  
 C) Decreases away from conductor  
 D) Decreases and then increases towards conductor
- Q.24 Magnetic Resonance Imaging (MRI) is used to identify the image of  
 A) Tumors and inflamed tissues ✓  
 B) Skin cells  
 C) Blood cells  
 D) Bone structures
- Q.25 Magnetic field strength is measure in:  
 A)  $\text{Wbm}^{-1}$   
 B)  $\text{Wbm}^2$   
 C)  $\text{Wbm}^{-2}$  ✓  
 D)  $\text{Wb}$
- Q.26 Force on current carrying conductor per unit length is given by:  
 A)  $IL \sin \theta$   
 C)  $IL$   
 B)  $IB \sin \theta$  ✓  
 D)  $IB \sin \theta$
- Q.27 If 'A' is fundamental dimension of ampere then the dimension of magnetic strength is:  
 A)  $[\text{MT}^2\text{A}^{-2}]$   
 B)  $[\text{MT}^2\text{L}^2\text{A}^{-1}]$   
 C)  $[\text{MT}^2\text{A}^{-1}]$  ✓  
 D)  $[\text{MT}^2\text{L}^{-2}\text{A}^{-2}]$
- Q.28 'F' is maximum force acting on a conductor. Now if we change the direction conductor by making an angle of  $45^\circ$  with the magnetic field then the force become  
 A)  $\frac{F}{2}$   
 B)  $\frac{F}{\sqrt{2}}$  ✓  
 C)  $2F$   
 D)  $\sqrt{2}F$
- Q.29 If we doubled all the parameters of the force acting on current carrying conductor and  $\theta = 90^\circ$  then magnetic force becomes:  
 A) Half  
 B) Eight-times ✓  
 C) Double  
 D) Four-times
- Q.30 The force acting on current carrying conductor will be maximum if the angle between magnetic field and conductor is:  
 A)  $0^\circ$   
 B)  $90^\circ$  ✓  
 C)  $30^\circ$   
 D)  $60^\circ$
- Q.31 Electric current is flowing through the circuit as shown in figure, what will be direction of magnetic lines of force:



- Q.32 A) Clockwise✓  
C) Anticlockwise  
The magnetic flux linked with a solenoid of area 'A', having 'N' turns at right angle to magnetic field is:  
A) NBA✓  
C) BA
- Q.33 B) From top to bottom  
D) From bottom to top  
A charge projected with velocity of 10m/s in a magnetic fields of 10T at an angle of 60°. If force exerted on charge is  $2.78 \times 10^{-17}$  N, then value of charge is:  
A)  $1.6 \times 10^{-19}$  C  
C)  $2.7 \times 10^{-19}$  C  
B)  $1/2 NBA$   
D)  $BA \cos(\theta)$
- Q.34 The value of magnetic flux is 20Wb, when magnetic lines of force containing magnetic field strength of 1T passing through unit area of  $10\text{m}^2$ , then angle between magnetic field and unit area is:  
A) 360°  
C) 180°✓  
B) 90°  
D) 45°
- Q.35 A loop of 5 turns of wire is placed in uniform magnetic field of 0.5T, then area of loop shrinks at a constant rate of  $10\text{m}^2/\text{s}$ , the emf induced is:  
A) 2.5 V  
C) 25V✓  
B) 250 V  
D) 0.25 V

## Modern Physics

- Q.1 Life time of electron in metastable state is about \_\_\_\_\_  
A)  $10^{-5}\text{sec}$   
C)  $10^{-3}\text{sec}$   
B)  $10^{-8}\text{sec}$ ✓  
D)  $10^{-2}\text{sec}$
- Q.2 The minimum frequency below which no electron is emitted from the metal surface is called  
A) High frequency  
C) Low frequency  
B) Threshold frequency✓  
D) Resonance frequency
- Q.3 In pair production, the type of photon used  
A)  $\alpha$ -particle  
C)  $\beta$ -particle  
B) X-rays  
D)  $\gamma$ -radiations✓
- Q.4 The life time of an electron in an excited state is about  $10^{-8}\text{s}$ . What is its uncertainty in energy during this time?  
A)  $1.05 \times 10^{-41}\text{J}$   
C)  $1.05 \times 10^{-26}\text{J}$ ✓  
B)  $1.15 \times 10^{10}\text{J}$   
D)  $2.19 \times 10^{-40}\text{J}$
- Q.5 Velocity of electron moving in first orbit of hydrogen is  
A)  $2.19 \times 10^7\text{m/sec}$   
C)  $2.18 \times 10^7\text{m/sec}$   
B)  $2.2 \times 10^8\text{m/sec}$   
D)  $2.19 \times 10^6\text{m/sec}$ ✓
- Q.6 LASER is a potential energy source for inducing which type of reaction?  
A) Radioactive  
C) Fission  
B) Ionization  
D) Fusion✓
- Q.7 Einstein's photoelectric equation is given by:  
A)  $hf = \phi + \frac{1}{2}mv^2$ ✓  
B)  $E = hc^2$



- B) Coherent monochromatic light is emitted  
 D) There is a population inversion of photons✓
- 21 What is meant by spontaneous emission of electrons in solids?  
 A) Electrons being emitted by the solids through photoelectric effect when irradiated with electromagnetic radiation  
 C) Incident electrons colliding with electrons in solids and releasing doubling the number of incident electrons  
 B) Electrons in solids are emitted without any external stimulus through radiation  
 D) Excited electrons going back to lower energy states immediately by releasing energy. ✓
- 22 When electrons lose all their kinetic energy in the first collision, the entire kinetic energy appears as an X-ray photon of energy:  
 A)  $K.E = eV$ ✓  
 B)  $K.E = \frac{hc}{\lambda_{min}}$   
 C)  $K.E = \frac{\lambda_{min}}{hc}$   
 D)  $K.E = \frac{hc}{\lambda_{max}}$
- 23 The characteristic X-ray spectrum is due to:  
 A) The absorption of neutrons by target material  
 C) The bombardment of target material by protons  
 B) The bombardment of target material by electrons✓  
 D) The bombardment of target material by alpha particles
- 24 The kinetic energy K.E. with which the electron strikes the target is given by:  
 A)  $K.E. = e^2V$   
 B)  $K.E. = hf^2$   
 C)  $K.E. = hc/\lambda$   
 D)  $K.E. = eV$ ✓
- 25 LASER is an acronym for:  
 A) Light amplification by stimulated emission of radiation✓  
 C) Light annihilation by stimulated emission of radiation  
 B) Light amplitude of stimulated emission of radiation  
 D) Light amplification by stimulated emission of radio
- 26 X-rays can be produced by bombardment of \_\_\_\_\_ on target metal:  
 A) Protons  
 B) Neutrons  
 C) Electrons✓  
 D) Alpha particles
- 27 Laser light is monochromatic which means  
 A) It consists of one ray of light  
 C) It consists of one wavelength✓  
 B) It consists of carbon monoxide gas  
 D) It consists of photons having 1 eV energy
- 28 If an electron in the 'K' shell is removed and an electron from 'L' shell jumps to occupy the hole in the 'K' shell, it emits a photon of energy:  
 A)  $hf_{K\alpha} = E_L - E_K$ ✓  
 B)  $h/\lambda_{K\alpha} = E_L - E_K$   
 C)  $hc = E_L - E_K$   
 D)  $hf_{K\alpha} = E_K - E_L$
- 29 Which of the following property must be there in a substance so that it can be used as target in X-ray tube?  
 A) It must have low melting point  
 B) It must have high reflecting ability  
 C) It must have low atomic number  
 D) It must have high atomic number✓
- 30 Which of the following can be used to produce population inversion for the emission



of Laser?

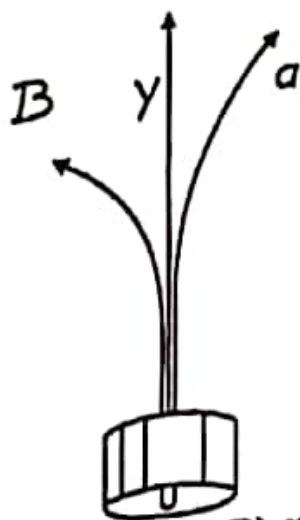
- A) Optical pumping ✓
- C) Optical fibre

- B) Optical instrument
- D) Optical polarization

## Nuclear Physics

- Q.1 In the half-life of an element, the equation for the number of decaying atoms is given by
- A)  $\Delta \propto -N\Delta t$  ✓
  - C)  $\Delta N = KN\Delta t$
  - B)  $\Delta N \propto -n\Delta t$
  - D)  $\Delta N = -\Delta N\Delta t$
- Q.2 Decay constant ' $\lambda$ ' is given as
- A)  $-\frac{\Delta N/N}{\Delta t}$  ✓
  - C)  $-\frac{\Delta N}{\Delta t}$
  - B)  $-\frac{N}{\Delta t}$
  - D)  $-\frac{\Delta N/N}{\Delta t}$
- Q.3 The SI unit of absorbed dose 'D' i.e. radiation effect is Gray and one Gray is equal to
- A) kJ / mol
  - C) J / mol
  - B) kg / J
  - D) J / kg ✓
- Q.4 The emission of  $\gamma$ -radiations from the nucleus is generally represented by the equation:
- A)  ${}_Z^AX \rightarrow {}_Z^AX^* + \gamma$  - radiations
  - C)  ${}_Z^AX^* \rightarrow {}_Z^AX + \beta$  - particles
  - B)  ${}_Z^AX^* \rightarrow {}_{Z-1}^AX + \gamma$  - radiations
  - D)  ${}_Z^AX^* \rightarrow {}_Z^AX + \gamma$  - radiations ✓
- Q.5 For intermediate energy of radiations, the dormant process is:
- A) Compton Effect ✓
  - C) Nuclear Effect
  - B) Photoelectric Effect
  - D) Pair Production
- Q.6 Ultraviolet radiations cause:
- A) Severe Crop Damage
  - C) Sunburn, blindness, skin cancer
  - B) Decay of Microorganisms
  - D) All of the above ✓
- Q.7 When a helium atom loses an electron, it becomes:
- A) An alpha particle
  - C) Proton.
  - B) A positive helium ion. ✓
  - D) A negative helium ion.
- Q.8 Beta ray emitted by a radioactive substance is:
- A) An electron which was existing outside the nucleus.
  - C) An electron which was existing inside the nucleus. ✓
  - B) An electron emitted by the nucleus as a result of the decay of neutron inside the nucleus.
  - D) A pulse of electromagnetic wave.
- Q.9 Which one is most stable element on the basis of binding energy?
- A) Sn.
  - C) Ba.
  - B) Kr.
  - D) Fe. ✓
- Q.10 \_\_\_\_\_ are the particles that experience strong nuclear force.
- A) Electrons.
  - B) Neutrinos.

- 11) C) Muons.  
The neutron is assumed to be made of  
A) One up quark and two down quarks.✓  
B) Two up quarks and one down quark.  
C) Two up quarks and two down quarks.  
D) One up quark and one down quark.
- 12) The cobalt is absorbed by  
A) Bones.  
C) Skin.  
B) Liver.✓  
D) Thyroid gland.
- 13) Ionizing capability of gamma rays is:  
A) Equal to alpha and beta particle  
C) Less than alpha but greater than beta particles  
B) Less than both alpha and beta particles✓  
D) Less than beta but greater than alpha particles
- 14) Half-life of a radioactive element is:  
A) Inversely proportional to square of decay constant  
C) Directly proportional to square of decay constant  
B) Directly proportional to decay constant  
D) Inversely proportional to decay constant✓
- 15) The transformation of a neutron into proton in the nucleus gives rise to emission of:  
A) Beta particles✓  
C) Alpha particles  
B) Gamma particles  
D) X-rays
- 16) The ratio of the rate of decay of a parent atom to the number of radioactive nuclei present at that time is equal to:  
A) Half-life of radioactive element  
C) Mean life  
B) Decay constant of radioactive element✓  
D) Activity of radioactive element
- 17) In a radioactive phenomenon observation shown in figure where  $\alpha$  deviates lesser than  $\beta$  in some electric or magnetic field (not shown in figure). What is the reason of less deviation of  $\alpha$ ?



- A)  $\alpha$  is charged particle  
C)  $\alpha$  is neutral particle

- B)  $\alpha$  is heavier particles✓  
D)  $\alpha$  is lighter particle

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